HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · ENERGY · FINANCE & ASSET OPERATIONS

RENEWABLES OPERATOR, INDIA · WIND & SOLAR · EIGHT AUTOMATIONS

Eight Chores, Automated

HMC automated eight workflows across generation, treasury, regulatory and lease accounting, matching each chore to the lightest tool that works: Excel formulas, VBA, Power Query or Python.

~~192~~ → **33** manual hours a month across the eight chores



8 chores
Workflows automated
across 4 toolkits



~159 hrs / month
Hours returned to the team
▼ from about 192 manual hours



83 %
Less manual handling
copy-paste, re-keying, lookups

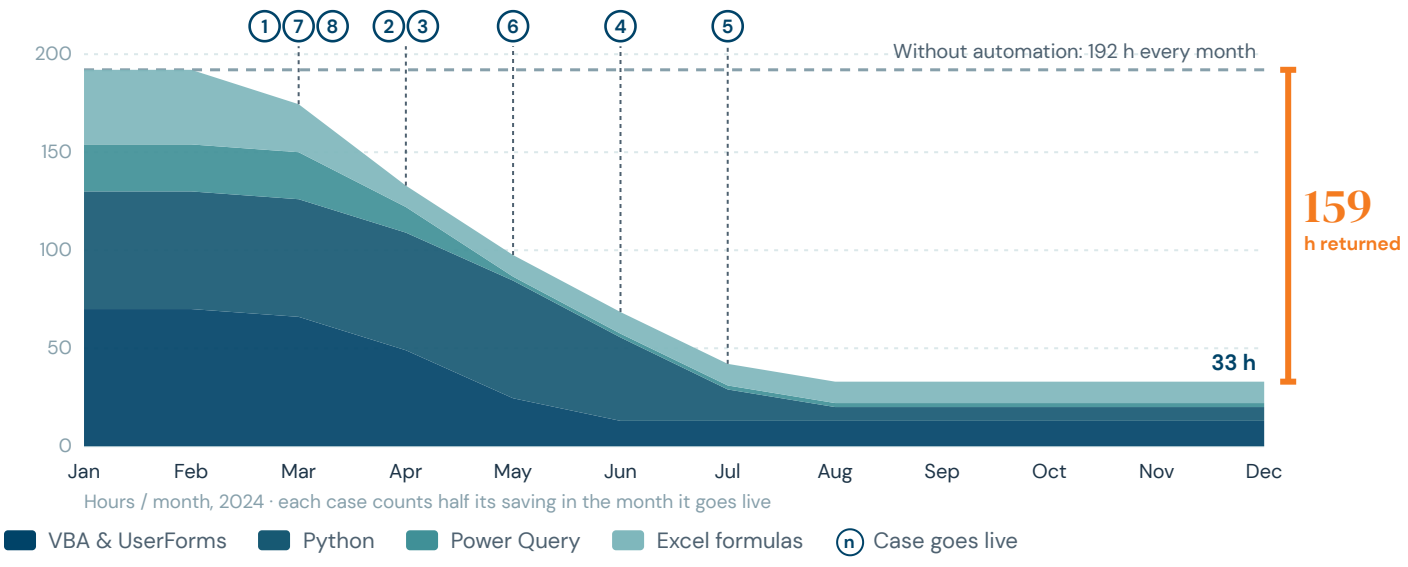


4 teams
Teams served
Generation · Treasury · Regulatory · Lease

Rings: one segment per chore, shaded by toolkit; 159 of 192 manual hours returned; share of manual handling removed; hours saved split by team. Figures indicative; validated metrics available on request.

01 Small automations, compounding hours

Manual hours per month in 2024, stacked by toolkit as each case went live



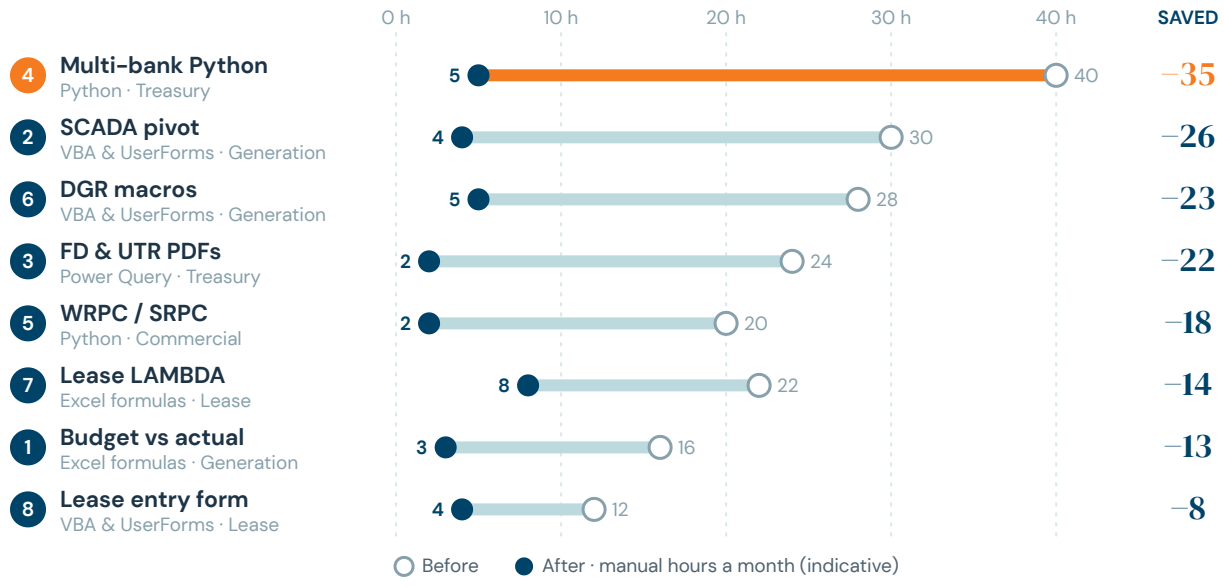
START
On-site week in Pune, March 2024
Sitting with each team to log every repeated task, how often it ran and how long it took.

ROLLOUT
Eight tools live March to July
Three cases shipped in March; the last, the regional energy-account extractor, in July.

RESULT
About four working weeks a month back
From 192 manual hours a month to 33, once all eight were running.

02 Eight chores, ranked by hours saved

Before → after, manual hours a month per case



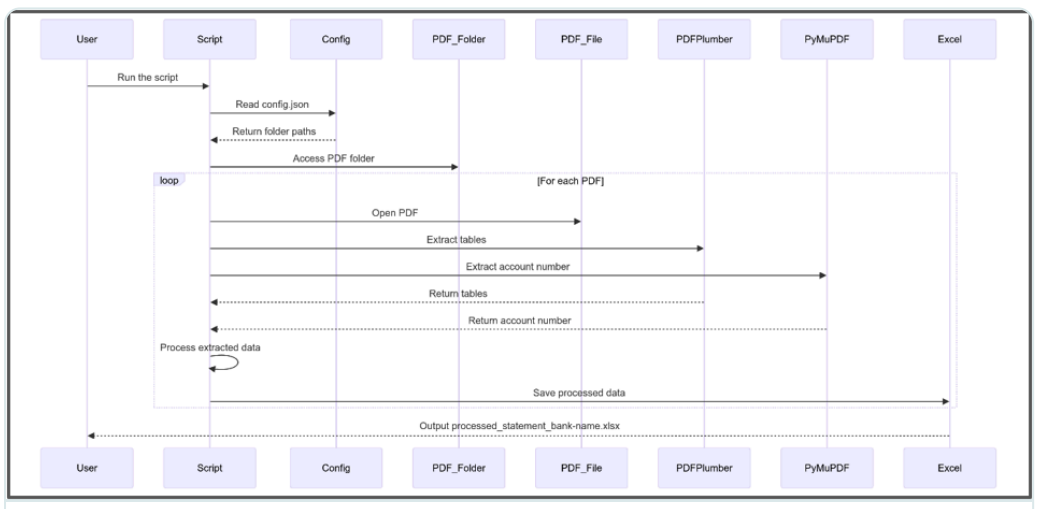
BIGGEST SINGLE WIN
Multi-bank consolidation in Python
Statements from every account pulled into one Excel file with regex checks: 40 hours a month down to 5.

NO SILVER BULLET
None saved more than 35 hours
The value came from the portfolio: eight modest tools, each owned by the team that runs it.

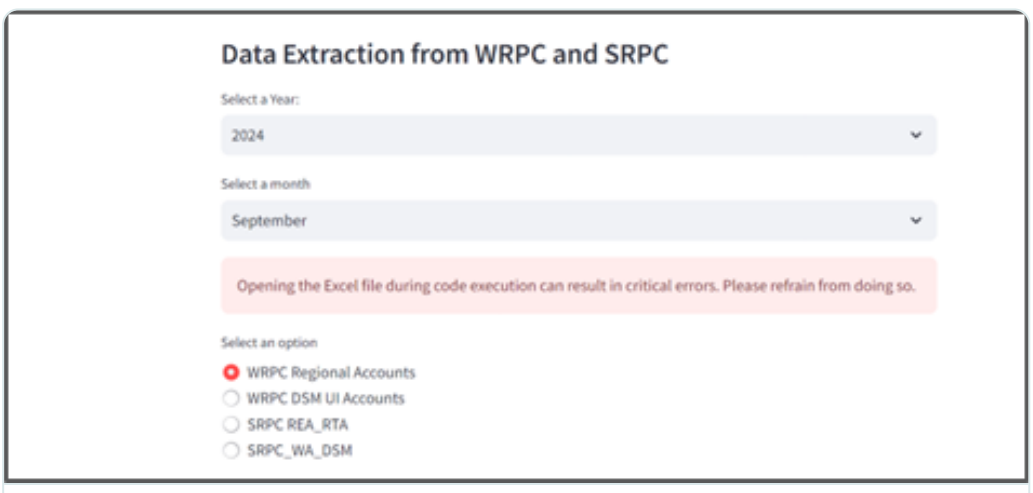
LIGHTEST TOOL FIRST
Python only where nothing else fits
Two of eight cases needed it: websites and messy PDFs. The rest are formulas, macros and Power Query.

03 The tools, as delivered

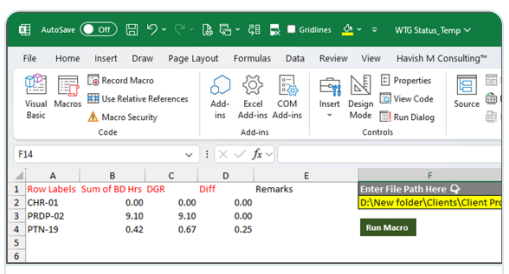
Original screens from the project · sensitive fields masked



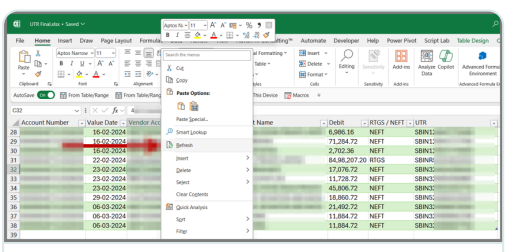
1 **Multi-bank pipeline**: config → PDF folder → pdfplumber / PyMuPDF → one Excel output per bank



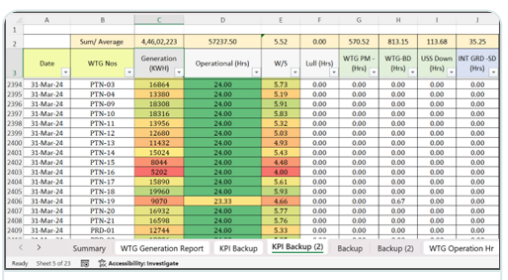
2 **Energy-account extractor**: pick year, month and account; Excel fills itself



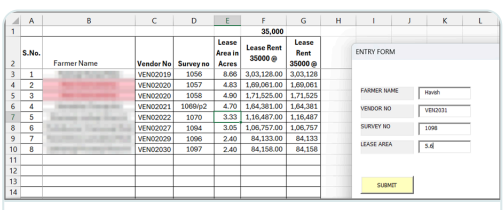
3 **SCADA macro**: BD hours vs DGR in one click



4 **FD & UTR table**: drop PDFs in a folder, Refresh



5 **DGR & KPI macros**: generation by turbine



6 **Lease entry form**: auto serial and vendor numbers

04 Four pain points, four fixes

Before → after

BEFORE
Data locked in PDFs
Bank statements and regional energy accounts read by eye and typed in.

POWER QUERY · PYTHON

AFTER
Folder, then Refresh
Drop PDFs in a folder and refresh, or run one script for all banks.

BEFORE
Copy, paste, repeat
SCADA and DGR data copied between workbooks by hand every day.

VBA MACROS

AFTER
One macro
Paste, filter, pivot and verify against the DGR in a single click.

BEFORE
Heavy, fragile workbooks
Lease models with long nested formulas over 14,000 rows, slow to recalculate.

DYNAMIC ARRAYS · LAMBDA

AFTER
Lean lease models
Dynamic arrays, named ranges and shared LAMBDA functions.

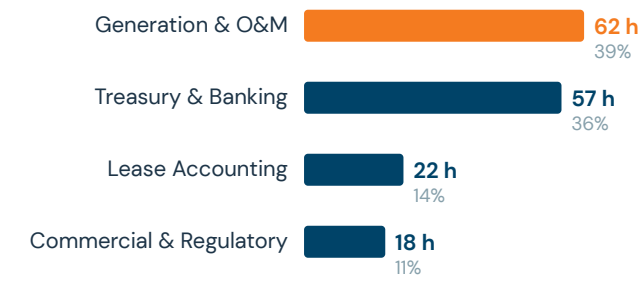
BEFORE
Mixed Excel skills
Some users wrote macros, others avoided formulas; entries had duplicates and gaps.

BUTTONS · USERFORMS

AFTER
A form anyone can use
Auto numbering, duplicate flags and a Submit button straight into the register.

05 Hours back, by team

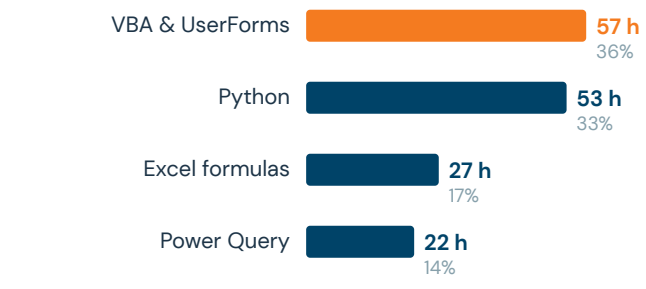
Generation & O&M gained the most



Hours saved per month · total 159 h

06 Hours back, by toolkit

Macros and UserForms did the heaviest lifting



Hours saved per month · total 159 h

07 How HMC ran it

1 week — Discover
On site beside the analysts; chore inventory ranked by hours and error risk.

1 per chore — Design
The lightest tool that fits; a design note signed off by each owner.

1–3 weeks each — Build
Tested on real files against last month's manual output.

✓ training — Adopt
Walkthrough, one-page guide and a support window per tool.

∞ ongoing — Scale
Site and month as parameters; new banks mapped once.

What any operations team can borrow

01 — Pick the lightest tool that works
A formula beat a macro, and a macro beat a new system. Python came in only where websites and messy PDFs left no other option.

02 — Small automations compound
None of the eight saved more than 35 hours a month on its own. Together they returned about four working weeks a month.

03 — Hand over with training
Every tool shipped with a walkthrough and a one-page guide, so the people who run the chore own it.

Built with **Excel 365** · **VBA** · **Power Query** · **Python** (pandas, pdfplumber, PyMuPDF) | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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